MIL-P-46427 (Ord)
16 November 1960

MILITARY SPECIFICATION

PRIMER, ELECTRIC, T104E9
METAL PARTS

1. SCOPE

1.1 This specification covers the metal parts for one type primer designated as Primer, Electric, T104E9, for use in artillery ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-G-2550 - General Specification for Ammunition Except Small Arms Ammunition.
MIL-P-10025 - Packing and Marking for Domestic Shipment of Inert Ammunition Components; General Specification for:

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.

DRAWINGS

ORDNANCE CORPS

CXP-97658 - Body
CXP-108349 - Retainer
CXP-108362 - Head

FSC 1390

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PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.
IEL-(see 6.3) - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Metal parts.-The metal parts shall comply with all requirements specified on Drawings (dwgs.) CXP-97658, CXP-108349, CXP-108362, as applicable, and with all requirements specified in applicable specifications.

3.3 Workmanship.-All parts shall be dry and free of chips, burrs, dust, grease, rust and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. Inspection shall be performed in accordance with this specification and other specifications referenced in any of the contractual documents.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies procured under this specification and shall perform all inspections required by the system. A current written description of the system shall be submitted to the contracting officer, and meet his approval, prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling

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procedures, methods of inspection (gages, measuring and testing equipment), and provisions for control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the provisions of 4.3 and 4.4 and the other documents referenced herein. The contractor shall not be restricted to the inspection station nor the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that affects the degree of assurance required by this specification or other documents referenced herein.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of analyses on all material procured directly by the contractor.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification shall consist of: (a) surveillance of contractor operations to determine that the methods and procedures of the written procedure are being properly applied and (b) Government product inspection. The Government inspector shall also ascertain prior to acceptance that the product complies with all the quality assurance provisions of other specifications referenced in any of the contractual documents.

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4.1.3.1 Surveillance.-Surveillance shall include, but is not limited to:

a. Observation of procedures concerning lot formation and identification.

b. Observation of sampling procedures and application of acceptance criteria.

c. Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.

d. Review of procedures for control and disposition of non-conforming material.

4.1.3.2 Product inspection.-Physical inspection by the Government of contractor accepted product will be performed to the extent necessary to assure contractor compliance with the requirements of this specification and to determine the reliability of contractor records. Severity of Government inspection of individual characteristics will be directly related to the seriousness classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records, Specification MIL-G-2550 will apply.

4.1.3.3 Deviations and deficiencies.-Deviations from the prescribed or agreed upon procedures, or instances of poor practice which might have an effect on the quality of the product, shall be called to the attention of the contractor immediately. To avoid interference with operation, the contractor shall designate a responsible official, or officials, to whom the Government inspector will report such instances. Failure of the contractor to correct deficiencies promptly shall be cause for suspension of acceptance until correction has been made or until conformance to prescribed criteria has been demonstrated.

4.2 Preproduction inspection.

4.2.1 Submission.-The contractor shall submit 50 each consecutively produced heads, bodies and retainers, which have been produced by the production process which the contractor will use in fulfilling the contract, to a Government-approved facility designated by the contracting officer. The preproduction quantity shall be submitted prior to initiation of production and whenever a change occurs in manufacturing process, material used, drawing, or specification, such as to significantly affect product uniformity as determined by the Government inspector.

4.2.2 Inspections to be performed.-Each component shall be inspected for all the inspections specified in 4.3 and 4.4 of this specification.

4.2.3 Rejection.-If any part fails to comply with the applicable requirements, the preproduction quantity shall be rejected and the contractor shall take the necessary corrective action and submit a new preproduction quantity, or portion thereof, as specified by the contracting officer. The contractor shall continue to submit new or supplemental quantities as necessary, until such time as a quantity successfully passes the inspection specified or until otherwise directed by the procuring activity.

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4.3 Inspection provisions (for production).

4.3.1 Lot formation.-A lot shall consist of metal parts produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, or fit, as determined by the Government, shall not necessitate changing the lot number.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Head (see dwg. CXP-108362).

Categories	Defects	Method of inspection
Critical:		
1.	Crack or split	Visual
Major:	AQL 1.50 percent	
101.	Pitch diameter, external thread	Gage
102.	Pitch diameter, internal thread	Gage
103.	Major diameter, external thread, minimum (min.)	Gage
104.	Minor diameter of internal thread, maximum (max.)	Gage
105.	Thickness of flange	Gage
106.	Concentricity of pitch diameters of internal and external threads	Gage
107.	Concentricity of electrode hole with counterbore	Gage
108.	Length of internal threads, min	Gage
109.	Length to external threads from end	Gage
110.	Diameter of electrode hole, min	Gage
111.	Diameter of electrode hole counterbore, min	Gage
112.	Thickness through bottom of electrode counterbore	Gage
113.	Depth to electrode hole counterbore	Gage
114.	Location of wrench holes	Gage
115.	Outside diameter of unthreaded area	Gage
116.	Concentricity between pitch diameter of internal thread with electrode hole counterbore	Gage

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Minor:	AQL 4.00 percent	
201.	Diameter of flange	Gage
202.	Diameter of electrode hole, max	Gage
203.	Diameter of electrode hole counterbore, max	Gage
204.	Diameter of wrench holes, min	Gage
205.	Depth of wrench holes, min	Gage
206.	Total length	Gage
207.	Width of external thread undercut, max	Gage
208.	Finish improper	Visual
209.	Evidence of poor workmanship (see 3.3)	Visual

4.3.2.2 Body (see dwg. CXP-97658).

Categories	Defects	Method of inspection
Critical:		
1.	Flash hole missing	Gage
2.	Crack or split	Visual
Major:	AQL 0.65 percent	
101.	Wall thickness, min	Gage
102.	Concentricity of outside diameter with pitch diameter of thread for min. required length	Gage
103.	Required plug gage fails to pass through body	Gage
104.	Finished body fails to pass through required ring gage	Gage
105.	Pitch diameter of thread	Gage
106.	Major diameter of thread, min	Gage
107.	Length of thread, min	Gage
108.	Protective coating damaged (base metal exposed)	Visual
Minor:	AQL 2.50 percent	
201.	Wall thickness, max	Gage
202.	Location of first pair of holes at unthreaded end	Gage
203.	Diameter of flash hole(s)	Gage
204.	Location of second pair of holes	Gage
205.	Total length	Gage
206.	Finish improper	Visual
207.	Evidence of poor workmanship (see 3.3)	Visual

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4.3.2.3 Retainer (see dwg. CXP-108349).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.40 percent	
101.	Pitch diameter of threads	Gage
102.	Major diameter of threads	Gage
103.	Flash hole missing	Visual
Minor:	AQL 2.50 percent	
201.	Width of screw slot, min	Gage
202.	Depth of screw slot, min	Gage
203.	Diameter at 45-degree counterbore	Gage
204.	Diameter of flash hole	Gage
205.	Length	Gage
206.	Finish improper	Visual
207.	Evidence of poor workmanship (see 3.3)	Visual

4.3.3 Testing.

4.3.3.1 Hydrostatic test (see dwg. CXP-97658) Critical defect.-This test shall be conducted 100 percent. If any primer tubing fails to comply with the applicable requirement, it shall be classed defective and no primer body shall be fabricated therefrom. (See 6.2). The test shall be performed as specified in 4.4.1, using equipment in accordance with 4.3.4.

4.3.4 Inspection equipment.-For the performance of the examinations and tests specified in 4.3 and 4.4, when applicable, the contractor shall have available, and utilize correctly, all necessary gages, measuring and testing equipment. (see 6.3). When inspecting the contractor's inspection equipment, the Government inspector will determine that the contractor has available, and utilizes correctly, gages, measuring and testing equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. The contractor will have available a set of master gages, standards, and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the contractor and made available for review by the Government. The calibration of gages, standards, and instruments will be periodically checked by authorized Government personnel.

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4.4 Test methods and procedures.

4.4.1 Hydrostatic test.-Prior to machining, the primer tubing shall be tested for not less than five seconds at the min. hydrostatic pressure as calculated by the following formula:

$$P = \frac{2\,St}{D} \times .9$$

where:

P = test pressure in pounds per square inch (psi).
S = min. specified yield strength, psi.
t = specified min. wall thickness in inches.
D = specified max. outside diameter in inches.

Observation shall be made for rupture, or indication of flaws as evidenced by leaks, and for permanent deformation beyond drawing requirements.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.

5.1.1 Level C.-Packing and marking shall be in accordance with Specification MIL-P-10025.

5.2 Data cards.-Data card information shall be in accordance with Specification MIL-G-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.2 Hydrostatic test.-If the primer tubing procured under this specification has been previously subjected to the hydrostatic test specified in this specification, a second hydrostatic test will not be required.

6.3 Inspection equipment.-Inspection equipment designs were not available at the time this specification was prepared. Prior to procurement, the procuring agency should contact Picatinny Arsenal, Dover, New Jersey, ATTN: ORDBB-ND, to determine availability of inspection equipment designs.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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